

MATS 9.0 Simplex Application - Round 2

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Code: <https://github.com/karthikviswanathan/mats8submissions/tree/main/mess3xbloch>

Train a transformer, where the tokens of each training sequence correspond to the Cartesian product of tokens from: (1) a Mess3 Process, and (2) a Bloch Walk Process. Since Mess3 has a vocabulary of 3 tokens, and Bloch Walk has a vocabulary of 4 tokens, the process from which training sequences are sampled has a vocabulary of 12 tokens. For each sequence of this “multipartite” training data, the constituent Mess3 and Bloch Walk sequences are generated independently of each other (each retains its own temporal structure).

1 Pre-registered Prediction

Make a pre-registered prediction, or several, (honor code) as to what geometry the activations should take. You will not be penalized for getting this wrong! Still, we are interested in your intuitions here, and what you think the possibilities might be.

Prediction: Orthogonal Subspace Decomposition

The residual stream will encode a belief geometry corresponding to $\Delta^2 \times \text{Bloch slice}$, where Δ^2 is the 2-simplex representing Mess3 belief states and the Bloch slice is the x - z plane of the Bloch sphere representing Bloch Walk belief states¹.

While the high-dimensional residual stream need not explicitly concatenate these representations (i.e., the first $D/2$ dimensions for η_M and the next $D/2$ for η_B), I predict that:

1. There exist two orthogonal *linear* subspaces in the residual stream
2. One subspace captures the Mess3 geometry (recovering the 2-simplex via linear projection), while the other captures the Bloch Walk geometry
3. These subspaces are identifiable via linear regression or other linear decomposition methods (e.g., PCA)

Overall Geometry: Expected PCA Structure

To establish a baseline expectation, we performed PCA on the ground truth joint belief states (Appendix A, Figure 4). When the 4-dimensional joint belief space ($\Delta^2 \times \text{Bloch slice}$) is decomposed via PCA, PC1-2 predominantly capture Bloch Walk geometry, while PC3-4 predominantly capture Mess3 geometry.

Prediction: PCA on the final layer residual stream activations should reveal a similar decomposition, with two principal components predominantly capturing Bloch Walk geometry (exhibiting

¹We denote $\eta_M \in \Delta^2$ for Mess3 belief states and $\eta_B \in \mathbb{R}^2$ for Bloch Walk belief states.

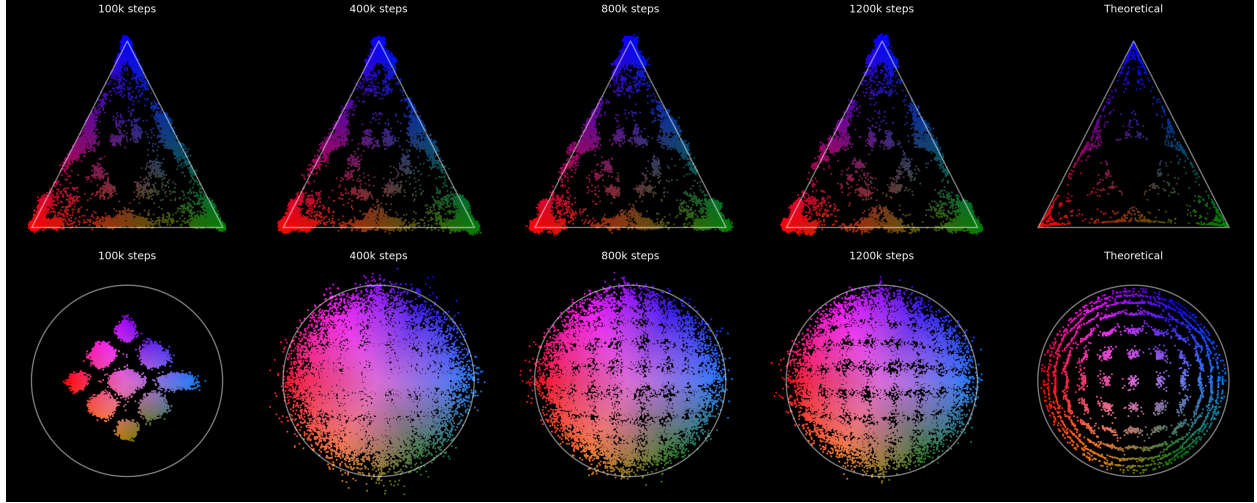


Figure 1: **Residual stream activations across training.** Columns show linear probe predictions from residual stream activations at checkpoints: 100k, 400k, 800k, 1.2M steps, and theoretical ground truth (rightmost) for the Mess3 process (top row) and the Bloch walk process (bottom row). We use 2000 sequences (16,000 activation samples) for this plot.

structure when colored by Bloch angle) and two capturing Mess3 geometry (exhibiting structure when colored by Mess3 belief state).

2 Training and Analysis of Residual Stream Activations

After training, analyze activations in the residual stream to determine if the belief state geometries of the Mess3 and Bloch Walk Process are in the residual stream. Are the individual geometries there? Tell us any explanations for why or why not?

Training

We trained a 4-layer transformer on the joint Mess3 $\times$ Bloch Walk process for 1.2M steps (60% of the planned 2M steps). Both belief state geometries emerge in the residual stream activations, as visualized in Figure 1.

Figure 5 shows the evolution of linear probe performance during training. Mess3 geometry emerges earlier, while Bloch Walk geometry develops progressively, with R^2 scores improving from 0.3 at initialization to 0.98 by 1.2M steps. RMSE decreases correspondingly, confirming that both belief geometries become increasingly accessible via linear readout throughout training.

At the final checkpoint, we observe high R^2 scores (0.99 for Mess3, 0.98 for Bloch Walk) and low RMSE (0.02 and 0.04 respectively), indicating that both belief geometries are linearly encoded in the residual stream.

Why Both Geometries Are Present

The next-token prediction factorizes over the joint process:

$$P(x_M, x_B | \text{history}) = P(x_M | \text{history}_M) \cdot P(x_B | \text{history}_B)$$

Since the two processes share no mutual information and prior work [4, 3] demonstrates that residual streams encode belief states for Mess3 and Bloch Walk processes individually, it is natural to expect the jointly-trained model to encode both belief states. One possible implementation would concatenate representations from individually-trained models, though the actual learned solution need not follow this structure.

3 Overall Geometry and Representation

Perform some analysis of your choosing, that speaks to the overall geometry and representation that the activations take in the residual stream. Please share any conclusions, thoughts, next directions etc. that you have.

PCA Analysis of Residual Stream Activations

To test our orthogonal subspace decomposition prediction (Section 1), we performed PCA on residual stream activations from the final checkpoint (1.2M steps). PC1-2 explain 93% of variance (51% and 42% respectively) and predominantly capture Mess3 geometry, while PC3-4 explain 3% of variance (1.5% and 1.4% respectively) and predominantly capture Bloch Walk geometry.

This pattern differs from the ground truth PCA baseline (Appendix A), where Bloch Walk dominated PC1-2.²

Figure 2 compares PCA projections (left column) with theoretical geometries (right column). The top row shows PC1 vs PC2 (colored by Mess3 belief state) alongside the theoretical Mess3 simplex; the bottom row shows PC3 vs PC4 (colored by Bloch angle) alongside the theoretical Bloch plane. Despite capturing only 3% of variance, the Bloch Walk geometry in PC3-4 is clearly present, confirming that both geometries are learned and occupy orthogonal subspaces with asymmetric variance allocation.

Despite capturing only 3% of variance, the Bloch Walk geometry in PC3-4 is clearly present, confirming that both geometries are learned and occupy orthogonal subspaces. The asymmetric variance allocation (93% to Mess3 versus 3% to Bloch Walk) is particularly striking given that both processes contribute equally to the next-token prediction task.

Conclusions, Thoughts and Next Directions

A striking observation from training is that learning the Bloch Walk geometry requires approximately 10 times more training steps when jointly trained with Mess3 compared to training on Bloch Walk alone. This substantial increase likely reflects the compounded challenge of simultaneously discovering two independent belief geometries while maintaining their orthogonality in the residual stream. Understanding why the model allocates variance so asymmetrically between the two geometries (93% to Mess3, 3% to Bloch) remains an open question worthy of investigation.

The phenomenon hints at a potential connection to superposition in neural networks [1], where features compete for limited representational capacity. Exploring this connection using the *technology* developed for quantum information theory could prove particularly fruitful and may offer novel insights into how neural networks manage compositional representations.

Future directions include:

²This discrepancy is expected: the ground truth PCA is merely a heuristic baseline on concatenated belief states, while the learned representation reflects how the network organizes information in the residual stream.

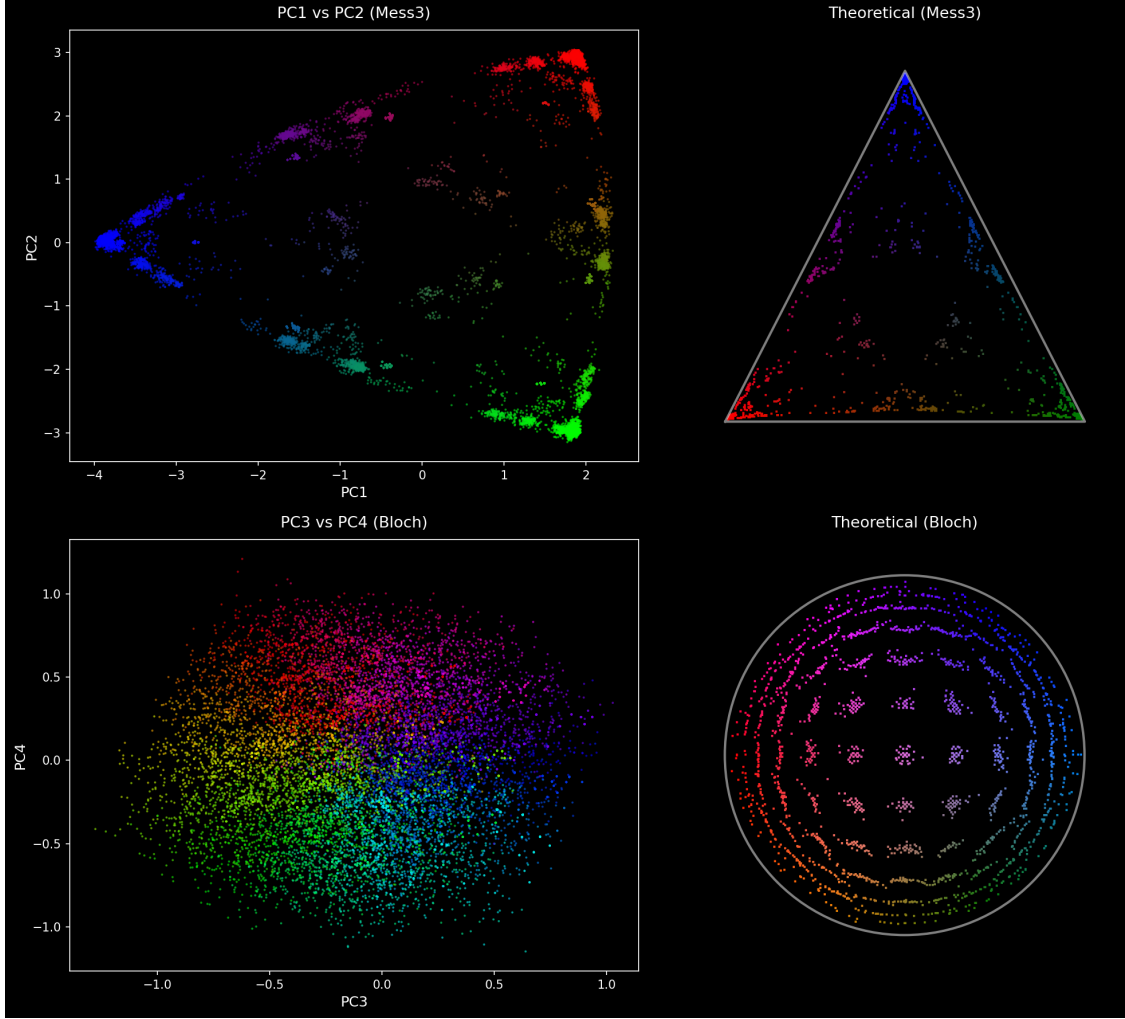


Figure 2: **PCA of residual stream activations compared with theoretical geometries.** **Top row:** PC1 vs PC2 (93% variance, colored by Mess3 belief state) and theoretical Mess3 simplex. **Bottom row:** PC3 vs PC4 (3% variance, colored by Bloch angle) and theoretical Bloch plane. PC1-2 predominantly capture Mess3 geometry while PC3-4 capture Bloch Walk geometry. Despite low variance, Bloch reconstruction is weak but present, confirming orthogonal subspace decomposition. We use 2000 sequences (16,000 activation samples) from the final checkpoint (1.2M steps).

- A mechanistic analysis [2] of how the model dynamically organizes these orthogonal subspaces during training.
- Investigating whether similar orthogonal decompositions emerge in natural language tasks with compositional structure.

4 Training SAEs on Residual Stream Activations

Extra Credit: Train an SAE on activations in the residual stream. What does the SAE learn? How does it relate to the belief state geometries of the Mess3 and Bloch Walk Processes?

Feature Specialization: Clean Orthogonal Decomposition

We trained a TopK SAE with $k=4$ and 512 learned features on residual stream activations from the final checkpoint. With extended training (50k steps, $\text{lr}=5 \times 10^{-4}$), the SAE achieves Bloch $R^2=0.946$. We found $k=4$ to be the smallest sparsity level achieving high R^2 (lower sparsity $k=2$ failed to recover Bloch geometry), demonstrating efficient factorized representations.

The $k=4$ SAE learns a reasonable factorization of the belief states (Figure 3):

- **3 pure Mess3 features:** Each monosemantically represents one of the three Mess3 states (selectivity > 2.3)
- **8 pure Bloch features:** Selectively encode Bloch x and z coordinates (correlation > 0.3)

This 3+8 decomposition aligns with the dimensionality of the two belief geometries: Mess3 requires 3 features (one per state on the 2-simplex), while Bloch Walk requires multiple features to represent the 2D Bloch plane geometry with sufficient fidelity, possibly due to the higher complexity of the Bloch walk geometry compared to the simplex.

Relationship to Residual Stream Geometry

The SAE’s learned features directly reflect the orthogonal subspace structure predicted in Section 1. This suggests the transformer internally maintains factorized representations of the two independent processes, consistent with the optimal solution for the joint prediction task.

Acknowledgements

Claude Code was used for coding and debugging assistance, and the SAE analysis was done during a vibe coding session. There may be some holes in the analysis, nevertheless the results looked neat enough to share for extra credit.

References

- [1] Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, Roger Grosse, Sam McCandlish, Jared Kaplan, Dario Amodei, Martin Wattenberg, and Christopher Olah. Toy models of superposition, 2022.
- [2] Mateusz Piotrowski, Paul M. Riechers, Daniel Filan, and Adam S. Shai. Constrained belief updates explain geometric structures in transformer representations, 2025.

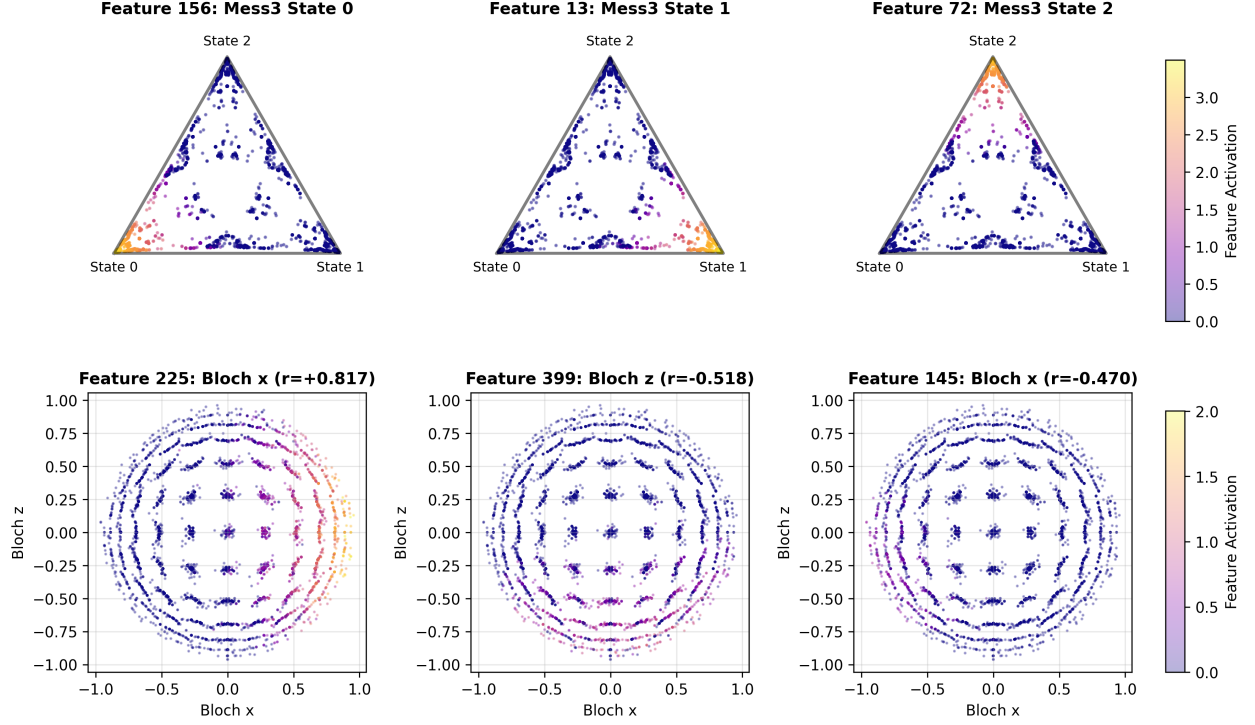


Figure 3: **TopK SAE ($k=4$) feature decomposition.** **Top row:** Three Mess3 features (156, 13, 72) plotted on the 2-simplex, each monosemantically representing one Mess3 state. **Bottom row:** Three Bloch-selective features showing strong correlations with Bloch x and z coordinates. The absence of mixed features (0 features selective for both processes) confirms perfect orthogonal separation.

- [3] Paul M. Riechers, Thomas J. Elliott, and Adam S. Shai. Neural networks leverage nominally quantum and post-quantum representations, 2025.
- [4] Adam S. Shai, Sarah E. Marzen, Lucas Teixeira, Alexander Gietelink Oldenziel, and Paul M. Riechers. Transformers represent belief state geometry in their residual stream. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, pages 75012–75034. Curran Associates, Inc., 2024.

A PCA Analysis of Ground Truth Joint Belief States

To establish a baseline expectation, we analyze the ground truth joint belief states using PCA. We concatenate the 2D barycentric projection of the Mess3 belief states with the 2D Bloch Walk belief states to form a 4-dimensional space representing $\Delta^2 \times \text{Bloch slice}$.

As shown in Figure 4, the PCA reveals a clear separation: PC1 and PC2 predominantly capture the Bloch Walk geometry, while PC3 and PC4 predominantly capture the Mess3 geometry. This demonstrates that the two independent processes naturally occupy orthogonal subspaces in the joint belief state space.

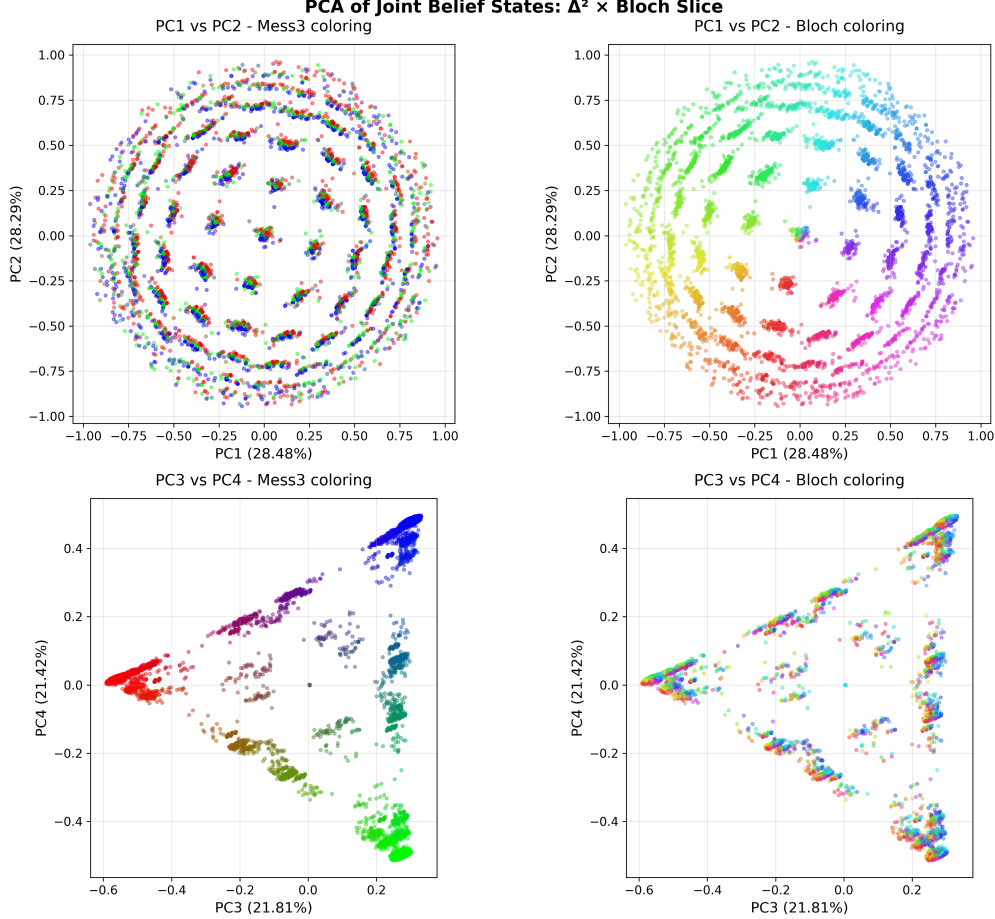


Figure 4: PCA of joint ground truth belief states. Top row: PC1 vs PC2. Bottom row: PC3 vs PC4. Left column: colored by Mess3 belief state. Right column: colored by Bloch Walk angle.

B Training Metrics: R^2 and RMSE Evolution

Figure 5 shows the evolution of linear probe performance during training. Both Mess3 and Bloch Walk geometries are progressively learned, with R^2 scores improving from near-zero at initialization to above 0.98 by 1.2M steps. The RMSE correspondingly decreases, confirming that the residual stream increasingly encodes both belief geometries in a linearly accessible form throughout training.

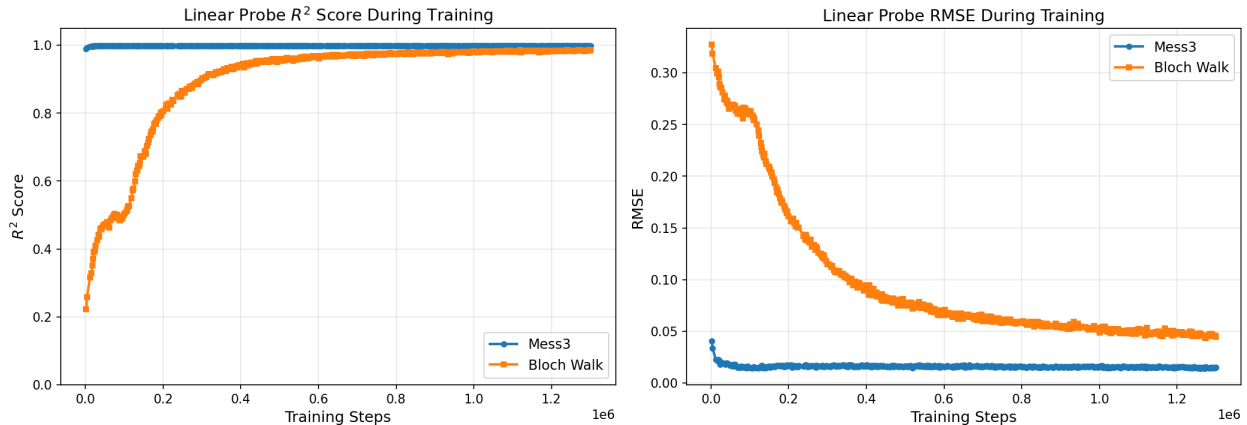


Figure 5: **Linear probe performance during training.** **Left:** R^2 scores for Mess3 (blue) and Bloch Walk (orange) linear probes during training. **Right:** RMSE for both processes. Both metrics show progressive improvement, confirming that belief state geometries emerge gradually and become increasingly linearly accessible in the residual stream as training progresses.

C Model Architecture and Training Hyperparameters

The transformer model used in this experiment was configured with the following architecture and trained with the hyperparameters listed below. Training was conducted for 1.2M steps (60% of the planned 2M steps) on the joint Mess3 $\times$ Bloch Walk process.

Parameter	Value
<i>Model Architecture</i>	
Number of layers	4
Model dimension (d_{model})	64
Number of attention heads	4
Head dimension (d_{head})	16
MLP hidden dimension (d_{mlp})	256
Context window length	8
Vocabulary size	12 (3×4)
Activation function	ReLU
<i>Training Hyperparameters</i>	
Total training steps	2,000,000 (planned)
Learning rate	10^{-4}
Batch size	128
Optimizer	Adam

Table 1: Architectural configuration and training hyperparameters for the joint process transformer.